



Inspiring Excellence

BRAC UNIVERSITY

Department of Computer Science and Engineering

MODULE HANDBOOK

Bachelor of Science in Computer Science (B.Sc. CS)

Programme:	B.Sc. in Computer Science
Department:	Computer Science and Engineering (CSE)
Faculty:	School of Data and Sciences
Duration:	4 Years (12 Trimesters)
Total Credits:	124 Credit Hours
Programme Code:	CS
Credit System:	Open Credit — Trimester Based

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1. Programme Overview

The Bachelor of Science in Computer Science (B.Sc. CS) at BRAC University is a rigorous 124-credit undergraduate programme designed to provide students with a comprehensive foundation in computer science theory and practice. The programme spans four academic years across twelve trimesters and is offered by the Department of Computer Science and Engineering (CSE), one of the flagship departments of BRAC University.

Students build competencies across a broad spectrum of computing disciplines — from low-level systems programming and digital logic to high-level software engineering, artificial intelligence, computer networks, and database systems. The curriculum is thoughtfully balanced between theoretical rigour and applied, project-based learning, preparing graduates for careers in both industry and academia.

The programme follows an Outcome-Based Education (OBE) framework. All graduates demonstrate measurable competencies mapped to twelve Programme Outcomes (POs) and four Programme Education Objectives (PEOs).

1.1 Programme Highlights

- Total Credit Hours: 124 (typically completed in 12 trimesters over 4 years)
- Open Credit Trimester System: Students choose courses and electives flexibly each trimester
- Mandatory Final-Year Project / Internship / Thesis (CSE 400 — 4 credits)
- Access to specialised research laboratories (CVIS Lab, Thesis Lab)
- Student tutoring support available 24/7 (on-campus 8 am–5 pm; online until 1 am)
- Strong industry placements with global technology companies
- Pathway for graduate studies at leading universities worldwide

1.2 Programme at a Glance

Qualification Awarded	Bachelor of Science in Computer Science (B.Sc. CS)
Awarding Institution	BRAC University, Dhaka, Bangladesh
Department	Computer Science and Engineering (CSE)
Faculty / School	School of Data and Sciences
Programme Code	CS
Mode of Study	Full-time, Face-to-Face with Laboratory Components
Normal Duration	4 Years (12 Trimesters)
Total Credit Hours	124 Credit Hours
Credit System	Open Credit — Trimester Based
Language of Instruction	English

2. Department Vision and Mission

2.1 Vision

The vision of the CSE Department is to inspire and engage students to be creative, industry-ready critical problem solvers, and to produce world-class researchers and entrepreneurs who will create jobs and solutions to build a prosperous Bangladesh.

2.2 Mission

M1	To prepare individuals with professional competency who will be able to contribute to technological advancement.
M2	To develop graduates with state-of-the-art multidisciplinary and collaborative research experience in computer science and engineering.
M3	To produce graduates enriched with human values and leadership skills.

3. Programme Education Objectives (PEOs)

The Programme Education Objectives describe the expected accomplishments of graduates 3–5 years after completing the B.Sc. CS degree.

PEO 1	Prove professional competency with effective computer science and engineering solutions to real-life problems.
PEO 2	Demonstrate excellence in terms of knowledge, teamwork, and leadership skills while maintaining the highest level of ethical standards.
PEO 3	Engage in the sustainable development of society through entrepreneurial and research skills.
PEO 4	Enrich their knowledge, skills, and professional growth through lifelong learning, higher education, and professional training.

3.1 PEO to Mission Statement Mapping

The table below shows how each Programme Education Objective (PEO) maps to the department's Mission Statements (M1–M3).

Mission Statement	PEO 1	PEO 2	PEO 3	PEO 4
M1 — To prepare individuals with professional competency who will contribute to technological advancement.	✓			✓
M2 — To develop graduates with multidisciplinary and collaborative research experience in CSE.		✓	✓	
M3 — To produce graduates enriched with human values and leadership skills.	✓		✓	

4. Programme Outcomes (POs)

Upon successful completion of the B.Sc. CS programme, graduates will be able to demonstrate the following twelve Programme Outcomes:

PO	Description
PO1	Apply knowledge of mathematics, natural science, computer science and engineering fundamentals and specialization to solve complex engineering problems.
PO2	Identify, formulate, research literature and analyze complex CSE problems reaching substantiated conclusions using first principles of mathematics and sciences.
PO3	Design solutions for complex CSE problems that meet specified needs with appropriate consideration for public health, safety, cultural, societal, and environmental factors.
PO4	Conduct investigations of complex problems using research-based knowledge including design of experiments, data analysis, and synthesis of information.
PO5	Create, select and apply appropriate techniques, resources, and modern engineering and IT tools to complex CSE problems.
PO6	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues relevant to professional engineering practice.
PO7	Understand and evaluate the sustainability and impact of professional engineering work in societal and environmental contexts.
PO8	Apply ethical principles and commit to professional ethics and norms of engineering and computing practice.
PO9	Function effectively as an individual, and as a member or leader in diverse, multi-disciplinary teams.
PO10	Communicate effectively on complex activities with the CSE community and society at large through reports, presentations, and design documentation.
PO11	Demonstrate knowledge of engineering management principles and economic decision-making applicable in team and project settings.
PO12	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning.

4.1 Mapping Programme Outcomes with PEOs

Each Programme Outcome (PO) is mapped to one or more Programme Education Objectives (PEOs) as shown below.

PO	PEO 1	PEO 2	PEO 3	PEO 4
PO1	✓			
PO2	✓			
PO3	✓			
PO4	✓			
PO5	✓			
PO6			✓	
PO7			✓	
PO8			✓	
PO9		✓		
PO10		✓		
PO11		✓		
PO12				✓

4.2 Mapping Courses with Programme Outcomes

The table below maps each course in the programme to the Programme Outcomes (POs) it addresses. An X indicates the course contributes to that outcome.

Course Code	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CSE 110	X	X	X		X							
CSE 111	X		X	X	X							
CSE 220	X		X									
CSE 221	X		X		X							
CSE 230	X	X										
CSE 250	X	X	X						X			
CSE 251	X	X	X		X							
CSE 260	X		X	X	X					X		
CSE 320	X	X		X					X			
CSE 321	X	X		X	X				X			
CSE 330	X	X	X									
CSE 331	X	X	X	X								
CSE 340	X	X					X					
CSE 341	X				X							
CSE 350			X	X	X							
CSE 360	X		X		X	X						
CSE 370	X		X		X				X			
CSE 420	X		X	X	X							
CSE 421	X	X	X		X	X		X		X		
CSE 422	X	X	X	X	X	X			X			X
CSE 423	X				X				X			
CSE 460	X	X	X	X	X				X			
CSE 461	X	X	X		X							X
CSE 470	X	X	X	X							X	
CSE 471	X		X	X				X	X			X
CSE 400	X	X	X	X	X	X	X	X	X	X	X	X

5. Programme Structure

The B.Sc. CS programme requires a total of 124 credit hours distributed across five categories as follows:

Category	Credits
University Core (General Education)	39
School Core	12
Programme Core	48
Programme Electives	21
Project / Internship / Thesis	4
Total	124

5.1 University Core — General Education (39 Credits)

General Education (GenEd) courses are mandatory for all BRAC University students regardless of major. The 39 GenEd credits are organised into five thematic streams. Each stream has a fixed set of required courses plus an elective chosen from a designated list. After completing the minimum 11 required GenEd courses, students select 2 additional GenEd electives from Streams 2–5 or from any department outside CSE.

Stream 1: Writing Comprehension

Develops academic reading, writing, and communication skills in English. Students must complete ENG 091 (if required) and progress through ENG 101 and ENG 102. ENG 103 is required only for students who entered through the ENG 102 pathway.

Course Code	Course Title	Credits	Requirement
ENG 091	Foundation Course (if required)	0.0	Conditional
ENG 101	English Fundamentals	3.0	Required
ENG 102	English Composition I	3.0	Required
ENG 103*	Advanced Writing Skills and Presentation	3.0	Required*

* ENG 103 is required only for students who entered the university via ENG 102 (freshers pathway).

Stream 2: Mathematics and Natural Sciences

Builds quantitative and scientific reasoning skills. MAT 092 is a remedial course for students who need it. All CS students must take MAT 110, PHY 111, and STA 201. One optional science elective may be taken from CHE 101, BIO 101, or ENV 103.

Course Code	Course Title	Credits	Requirement
MAT 092	Remedial Course in Mathematics (if required)	0.0	Conditional
MAT 110	Mathematics I: Differential Calculus & Co-ordinate Geometry	3.0	Required
PHY 111	Principles of Physics I	3.0	Required
STA 201	Elements of Statistics and Probability	3.0	Required
CHE 101 / BIO 101 / ENV 103	Science Elective (choose one, optional)	3.0	Optional

Stream 3: Arts and Humanities

Cultivates ethical reasoning, cultural awareness, and literary appreciation. HUM 103 and BNG 103 are required for all students. Students must additionally choose at least one elective from the designated list:

HUM101, HUM102, HST102, HST103, HST104, HUM207, ENG110, ENG113, ENG114, ENG115, or ENG333.

Course Code	Course Title	Credits	Requirement
HUM 103	Ethics and Culture	3.0	Required
BNG 103	Bangla Language and Literature	3.0	Required
Arts & Humanities Elective	One from: HUM101, HUM102, HST102, HST103, HST104, HUM207, ENG110, ENG113, ENG114, ENG115, ENG333	3.0	Choose 1

Example: HUM 101 — Introduction to World Civilization and Culture.

Stream 4: Social Sciences

Develops understanding of political, economic, social, and historical contexts. EMB 101 / DEV 101 is required for all students. Students must additionally choose at least one elective from the designated social sciences list.

Course Code	Course Title	Credits	Requirement
EMB 101 / DEV 101	Emergence of Bangladesh / Bangladesh Studies	3.0	Required
Social Sciences Elective	One from: PSY101, SOC101, ANT101, POL101, BUS201, ECO101, ECO102, ECO105, BUS102, POL102–103, POL201–202, PSY102, DEV104, DEV201, SOC201/ANT202, ANT342, ANT351, BUS333–335	3.0	Choose 1

Example: POL 101 — Introduction to Political Science; ECO 101 — Introduction to Microeconomics; SOC 101 — Introduction to Sociology.

Stream 5: Community Seeking Transformation (CST)

Capstone interdisciplinary stream unique to BRAC University's GenEd framework. CST courses engage students on cross-disciplinary topics — including food and nutrition, wellbeing, peace, conflict and justice, religion, arts, history, and corporate anthropology — that transcend any single academic discipline. Students must choose at least one course from: CST201, CST301–CST310.

Course Code	Course Title	Credits	Requirement
CST Elective	One from: CST201, CST301, CST302, CST303, CST304, CST305, CST306, CST307, CST308, CST309, CST310	3.0	Choose 1

Examples: CST305 — Borders and Belonging: Past and Future; CST309 — Global Citizenship.

GenEd Free Electives (2 courses × 3 credits = 6 credits)

After completing the 11 minimum required GenEd courses across Streams 1–5 above, students choose any 2 additional elective courses from Streams 2, 3, 4, or 5, or from any department outside the CSE department. These are free choices — students may select any course that interests them from the eligible pool.

Total GenEd Credits: 39 (comprising required stream courses + 2 free electives)

5.2 School Core (12 Credits)

Course Code	Course Title	Credits	Type
MAT 120	Mathematics II: Integral Calculus and Differential Equations	3.0	Core
MAT 215	Mathematics III: Complex Variables and Laplace Transformations	3.0	Core
MAT 216	Mathematics IV: Linear Algebra and Fourier Analysis	3.0	Core
PHY 112	Principles of Physics II	3.0	Core

5.3 Programme Core (48 Credits)

Course Code	Course Title	Credits	Type
CSE 110	Programming Language I	3.0	Core

Course Code	Course Title	Credits	Type
CSE 111	Programming Language II	3.0	Core
CSE 220	Data Structures	3.0	Core
CSE 221	Algorithms	3.0	Core
CSE 230	Discrete Mathematics	3.0	Core
CSE 260	Digital Logic Design	3.0	Core
CSE 321	Operating Systems	3.0	Core
CSE 330	Numerical Methods	3.0	Core
CSE 331	Automata and Computability	3.0	Core
CSE 340	Computer Architecture	3.0	Core
CSE 370	Database Systems	3.0	Core
CSE 420	Compiler Design	3.0	Core
CSE 421	Computer Networks	3.0	Core
CSE 422	Artificial Intelligence	3.0	Core
CSE 423	Computer Graphics	3.0	Core
CSE 470	Software Engineering	3.0	Core
CSE 400	Final-Year Project / Internship / Thesis	4.0	Capstone

5.4 Programme Electives (Minimum 21 Credits)

Students must complete a minimum of 21 elective credits. At least one elective must be from the CSE department. The remaining credits may be drawn from CSE electives, other departments (COD), GenEd courses, or a declared Minor / Second Major.

Course Code	Course Title	Credits	Type
CSE 250	Circuits and Electronics	3.0	Elective
CSE 251	Electronic Devices and Circuits	3.0	Elective
CSE 310	Object Oriented Programming	3.0	Elective
CSE 320	Data Communications	3.0	Elective
CSE 341	Microprocessors	3.0	Elective
CSE 342	Computer Systems Engineering	3.0	Elective
CSE 350	Digital Electronics and Pulse Techniques	3.0	Elective
CSE 360	Computer Interfacing	3.0	Elective
CSE 390	Technical Communication	3.0	Elective
CSE 391	Programming for the Internet	3.0	Elective
CSE 392	Signals and Systems	3.0	Elective
CSE 410	Advanced Programming in UNIX	3.0	Elective
CSE 419	Programming Languages and Competitive Programming	3.0	Elective
CSE 424	Pattern Recognition	3.0	Elective
CSE 425	Neural Networks	3.0	Elective
CSE 426	Advanced Algorithms (Graph Theory)	3.0	Elective
CSE 427	Machine Learning	3.0	Elective
CSE 428	Image Processing	3.0	Elective

Course Code	Course Title	Credits	Type
CSE 429	Basic Multimedia Theory	3.0	Elective
CSE 430	Digital Signal Processing	3.0	Elective
CSE 431	Natural Language Processing	3.0	Elective
CSE 432	Speech Recognition and Synthesis	3.0	Elective
CSE 460	VLSI Design	3.0	Elective
CSE 461	Introduction to Robotics	3.0	Elective
CSE 462	Fault Tolerant Systems	3.0	Elective
CSE 471	System Analysis and Design	3.0	Elective
CSE 472	Human Computer Interface	3.0	Elective
CSE 473	Decision Support System	3.0	Elective
CSE 474	Simulation and Modeling	3.0	Elective
CSE 490	WAN Routing and Technologies / Special Topics	3.0	Elective
CSE 491	Independent Study	3.0	Elective
CSE 496	Ethical Hacking	3.0	Elective

6. Prerequisite Structure

BRAC University operates an open credit trimester-based system. Course prerequisites are classified as Hard Prerequisites (HP — must be completed before enrolment) or Soft Prerequisites (SP — recommended but not mandatory). The following table outlines key prerequisite chains in the programme.

Course	Direct Prerequisite	Full Prerequisite Chain
CSE 111	CSE 110 (HP)	CSE 110
CSE 220	CSE 111 (HP), CSE 230 (HP)	CSE 230 → CSE 111 → CSE 110
CSE 221	CSE 220 (HP)	CSE 220 → CSE 111 → CSE 110
CSE 310	CSE 370 (HP)	CSE 370 → CSE 221 → CSE 220 → CSE 111 → CSE 110
CSE 321	CSE 221 (HP)	CSE 221 → CSE 220 → CSE 111 → CSE 110
CSE 330	MAT 216 (HP)	MAT 216 → MAT 120 → MAT 110
CSE 331	CSE 221 (HP)	CSE 221 → CSE 220 → CSE 111 → CSE 110
CSE 340	CSE 260 (HP)	CSE 260
CSE 370	CSE 221 (HP)	CSE 221 → CSE 220 → CSE 111 → CSE 110
CSE 391	CSE 370 (HP)	CSE 370 → CSE 221 → CSE 220 → CSE 111 → CSE 110
CSE 410	CSE 321 (HP)	CSE 321 → CSE 221 → CSE 220 → CSE 111 → CSE 110
CSE 420	CSE 321, CSE 331, CSE 340 (HP)	CSE 321 / CSE 331 / CSE 340 → ... → CSE 110
CSE 422	CSE 221 (HP)	CSE 221 → CSE 220 → CSE 111 → CSE 110
CSE 423	MAT 216 (HP)	MAT 216 → MAT 120 → MAT 110
CSE 430	MAT 120 (HP)	MAT 120 → MAT 110
CSE 470	CSE 370 (HP)	CSE 370 → CSE 221 → CSE 220 → CSE 111 → CSE 110
MAT 120	MAT 110 (HP)	MAT 110
MAT 215	MAT 216 (HP)	MAT 216 → MAT 120 → MAT 110
MAT 216	MAT 120 (HP)	MAT 120 → MAT 110

7. Module / Course Descriptions

This section provides detailed descriptions of all Programme Core courses, key elective courses offered by the CSE Department, and supporting courses from Mathematics and Physics. Each entry specifies the course code, title, credit value, prerequisites, learning content, and recommended texts.

7.1 Programme Core Courses

CSE 110: Programming Language I (3 Credits)	
Course Code	CSE 110
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz + Assignments + Mid Term Exam + Lab + Final Exam
Course Description	An introduction to the foundations of computation and mechanized computation. Emphasis is placed on techniques of problem analysis and the development of algorithms and programs. Topics include: introduction to digital computers and programming; algorithm and flowchart construction; information representation in digital computers; writing, debugging and running programs (including file handling); data structures; abstraction; recursion; iteration; and the design and analysis of basic algorithms. The course includes a compulsory 3-hour laboratory session each week.
Suggested Reading	<i>Appropriate textbook for the chosen programming language (selected by the course instructor).</i>

CSE 111: Programming Language II (3 Credits)	
Course Code	CSE 111
Credits	3 Credit Hours
Prerequisite(s)	CSE 110 (Hard)
Assessment	Quiz + Assignments + Mid Term Exam + Lab + Final Exam
Course Description	An introduction to data structures, formal specification of syntax, elements of language theory, and mathematical preliminaries. Topics include formal languages, structured programming concepts, and a survey of features of existing high-level languages. Students design and write applications using an appropriate programming language. The course includes a compulsory 3-hour laboratory session each week.
Suggested Reading	<i>Appropriate textbook for the chosen programming language (selected by the course instructor).</i>

CSE 220: Data Structures (3 Credits)	
Course Code	CSE 220
Credits	3 Credit Hours
Prerequisite(s)	CSE 111 (Hard), CSE 230 (Hard)
Assessment	Quiz + Assignments + Mid Term Exam + Lab + Final Exam
Course Description	Introduction to widely used and effective methods of data organisation. Focuses on data structures, their algorithms, and algorithm performance. Topics include: elementary data objects; arrays, lists, stacks, queues, graphs, and trees;

	compound structures; data abstraction; primitive operations; memory management; sorting and searching; hash techniques; recursion; backtrack search; priority queues; graph dictionaries; and an introduction to database systems. Includes a compulsory 3-hour laboratory session on alternate weeks.
Suggested Reading	<i>Jeffrey Kingston, Algorithms and Data Structures: Design, Correctness, Analysis (2nd ed., Addison-Wesley). Aho, Hopcroft & Ullman, Data Structures and Algorithms (Addison-Wesley, 1974).</i>

CSE 221: Algorithms (3 Credits)

Course Code	CSE 221
Credits	3 Credit Hours
Prerequisite(s)	CSE 220 (Hard)
Assessment	Quiz + Assignments + Mid Term Exam + Lab + Final Exam
Course Description	The study of efficient algorithms and effective algorithm design techniques. Topics include: analysis techniques; Divide and Conquer; Greedy method; Dynamic Programming; Backtracking; basic search and traversal techniques; graph algorithms; elementary parallel algorithms; algebraic simplification; lower bound theory; NP-hard and NP-complete problems. Particular classes of algorithms, including sorting and searching, are studied in detail. Includes a compulsory 3-hour laboratory on alternate weeks.
Suggested Reading	<i>Donald E. Knuth, The Art of Computer Programming Vol. 1: Fundamental Algorithms (3rd ed., Addison-Wesley, 1997).</i>

CSE 230: Discrete Mathematics (3 Credits)

Course Code	CSE 230
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz + Assignments + Mid Term Exam + Final Exam
Course Description	Set theory; elementary number theory; graph theory; paths and trees; Boolean algebra; binary relations; functions; algebraic systems; generating functions; induction; reduction; semigroups; permutation groups; discrete probability; mathematical logic; propositional calculus; and predicate calculus.
Suggested Reading	<i>Kenneth H. Rosen, Discrete Mathematics and Its Applications (5th ed., McGraw-Hill, 2003). Bernard Kolman et al., Discrete Mathematical Structures (5th ed., Prentice Hall, 2003).</i>

CSE 260: Digital Logic Design (3 Credits)

Course Code	CSE 260
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz + Assignments + Mid Term Exam + Lab + Project + Final Exam
Course Description	An introduction to digital systems including computers, communication, and information systems. Topics: Boolean algebra; digital logic gates; combinational logic circuits; decoders, encoders, and multiplexers; asynchronous and synchronous counters; registers; flip-flops; adders; sequential circuit analysis and design; simple computer architecture. Includes a compulsory 3-hour laboratory each week.
Suggested Reading	<i>Ronald J. Tocci, Neal S. Widmer & Gregory L. Moss, Digital Systems:</i>

Principles and Applications (9th ed., Prentice Hall, 2003).

CSE 321: Operating Systems (3 Credits)

Course Code	CSE 321
Credits	3 Credit Hours
Prerequisite(s)	CSE 221 (Hard)
Assessment	Quiz + Assignments + Mid Term Exam + Lab + Final Exam
Course Description	Principles of operating systems: design objectives; sequential and concurrent processes; functional mutual exclusion; processor co-operation and deadlocks; management; control and scheduling of large information-processing systems; dispatching; processor access methods; job control languages; memory addressing; paging and store multiplexing; time sharing; batch processing; scheduling algorithms; file systems; security; semaphores and critical sections; device drivers; multiprocessing; sharing; design and implementation methodology; performance evaluation; and case studies. Includes a compulsory 3-hour laboratory each week.
Suggested Reading	<i>Silberschatz, Galvin & Gagne, Operating System Concepts (6th ed., Wiley, 2002).</i> <i>A. Tanenbaum, Modern Operating Systems (2nd ed., Prentice-Hall, 2001).</i>

CSE 330: Numerical Methods (3 Credits)

Course Code	CSE 330
Credits	3 Credit Hours
Prerequisite(s)	MAT 120 (Hard)
Assessment	Quiz + Assignments + Mid Term Exam + Lab + Final Exam
Course Description	Computer arithmetic; floating-point representation; arithmetic operations with normalised floating-point numbers; iterative methods for finding roots of equations and their computer implementation; solution of simultaneous algebraic equations (Gauss elimination); interpolation; least square approximation; Taylor and Chebyshev series; numerical differentiation and integration; numerical solution of differential equations.
Suggested Reading	<i>S.B. Rao & C.K. Shantha, Numerical Methods (Vantage Press, 1992).</i> <i>P. Balagurusamy, Numerical Methods (Techmedia).</i>

CSE 331: Automata and Computability (3 Credits)

Course Code	CSE 331
Credits	3 Credit Hours
Prerequisite(s)	CSE 221 (Hard)
Assessment	Quiz + Assignments + Mid Term Exam + Final Exam
Course Description	An introduction to finite representation of infinite objects and basic mathematical models of computation. Topics: finite automata and regular languages; pushdown automata and context-free languages; Turing machines; Church's Thesis; partial recursive functions; undecidability; reducibility and completeness; the Halting Problem; time complexity; NP-completeness; probabilistic computation; and interactive proof systems.
Suggested Reading	<i>H.R. Lewis & C.H. Papadimitriou, Elements of the Theory of Computation (2nd ed., Prentice Hall, 1997).</i> <i>J.V. Kohavi, Switching Theory and Finite Automata (McGraw-Hill, 1979).</i>

CSE 340: Computer Architecture (3 Credits)

Course Code	CSE 340
Credits	3 Credit Hours
Prerequisite(s)	CSE 260 (Hard)
Assessment	Quiz + Assignments + Mid Term Exam + Final Exam
Course Description	A systematic study of elements in computer design: circuit design; storage mechanisms; addressing schemes; various approaches to parallelism and distributed logic; information representation and transfer; instruction and data access methods; the control unit; hardware and microprogrammed control; memory organisation; RISC and CISC machines.
Suggested Reading	<i>Patterson, Hennessy et al., Computer Organization and Design: The Hardware/Software Interface (3rd ed., Morgan Kaufmann, 2004).</i>

CSE 370: Database Systems (3 Credits)

Course Code	CSE 370
Credits	3 Credit Hours
Prerequisite(s)	CSE 221 (Hard)
Assessment	Quiz + Assignments + Mid Term Exam + Lab + Project + Final Exam
Course Description	Introduction to concepts and methods for storing and manipulating data. Topics: file retrieval and organisation; database models and design; principles of database management systems (DBMS); relational DBMS; query formulation and SQL; database administration; storage, selection, and presentation of data; database integrity and security. Students complete four SQL lab assignments and work with popular database application packages.
Suggested Reading	<i>Elmasri & Navathe, Fundamentals of Database Systems (4th ed., Addison-Wesley, 2003). P. Bhattacharya, Database Management Systems (Tata McGraw-Hill).</i>

CSE 420: Compiler Design (3 Credits)

Course Code	CSE 420
Credits	3 Credit Hours
Prerequisite(s)	CSE 321 (Hard), CSE 331 (Hard), CSE 340 (Hard)
Assessment	Quiz + Assignments + Mid Term Exam + Lab + Final Exam
Course Description	Theory and practice of compiler and interpreter design, with emphasis on practical solutions using compiler-writing tools such as Yacc. Topics: lexical scanners; context-free languages and pushdown automata; recursive descent parsing; bottom-up parsing; attributed grammars; symbol table design; run-time memory allocation; machine language; code generation; and optimisation. Includes a compulsory 3-hour laboratory on alternate weeks.
Suggested Reading	<i>Aho & Ullman, Principles of Compiler Design (Addison-Wesley, 1977). Kenneth C. Loudon, Compiler Construction: Principles and Practice (1997).</i>

CSE 421: Computer Networks (3 Credits)

Course Code	CSE 421
Credits	3 Credit Hours

Prerequisite(s)	None (CSE 320 required for CCNA pathway)
Assessment	Quiz + Assignments + Mid Term Exam + Lab + Final Exam
Course Description	Transport connections and sessions; protocol hierarchy; design issues in transport and session layers; end-to-end protocols; message handling; terminal and file transfer protocols; Internet TCP/IP protocols; congestion control; wireless networks; mobile computing; high-speed networks; concurrent programming; data link layer; framing and error control; media access control; distributed computation; resource management; distributed file systems; fault tolerance; network security and privacy; deadlock detection; synchronisation; and concurrency control.
Suggested Reading	<i>U.D. Black, Computer Networks: Protocols, Standards and Interfaces (5th ed., Prentice Hall, 1987).</i>

CSE 422: Artificial Intelligence (3 Credits)

Course Code	CSE 422
Credits	3 Credit Hours
Prerequisite(s)	CSE 221 (Hard)
Assessment	Quiz + Assignments + Mid Term Exam + Lab + Final Exam
Course Description	Survey of artificial intelligence concepts: knowledge representation; search and control techniques; AI machines; LISP and PROLOG; problem representation; constraint propagation; rule chaining; frame inheritance; inference and learning in intelligent systems; systems for general problem solving; game playing; expert consultation; concept formation; natural language processing; recognition; understanding and translation; heuristic vs. algorithmic programming; cognitive simulation; machine intelligence; and expert systems (robotics). Includes compulsory 3-hour laboratory each week.
Suggested Reading	<i>Stuart J. Russell & Peter Norvig, Artificial Intelligence: A Modern Approach (2nd ed., Prentice Hall, 2002).</i>

CSE 423: Computer Graphics (3 Credits)

Course Code	CSE 423
Credits	3 Credit Hours
Prerequisite(s)	MAT 215 (Hard)
Assessment	Quiz + Assignments + Mid Term Exam + Lab + Project + Final Exam
Course Description	Introduction to graphical data processing; fundamentals of interactive graphics; architecture of display devices and connectivity to a computer; implementation of 2D and 3D viewing; clipping and transformations; hidden line algorithms; raster graphics concepts (architecture, algorithms, and image synthesis methods); design of interactive graphic conversations. Includes a compulsory 3-hour laboratory on alternate weeks.
Suggested Reading	<i>Newman & Sproull, Principles of Interactive Computer Graphics (2nd ed., McGraw-Hill, 1978). Foley, van Dam, Feiner & Hughes, Computer Graphics: Principle and Practice in C (2nd ed., Addison-Wesley, 1995).</i>

CSE 470: Software Engineering (3 Credits)

Course Code	CSE 470
Credits	3 Credit Hours
Prerequisite(s)	CSE 370 (Hard)

Assessment	Quiz + Assignments + Mid Term Exam + Project + Final Exam
Course Description	Concepts of software engineering: requirements definition; modular and structured design; data and functional specifications; verification; documentation; software maintenance; software support tools; software project organisation; quality assurance; management; and communication skills.
Suggested Reading	<i>Ian Sommerville, Software Engineering (6th ed., Addison-Wesley, 2000). Roger S. Pressman, Software Engineering: A Practitioner's Approach (6th ed., McGraw-Hill, 2004).</i>

CSE 400: Final-Year Project / Internship / Thesis (4 Credits)

Course Code	CSE 400
Credits	4 Credit Hours
Prerequisite(s)	Completion of required programme credits (see Department guidelines)
Assessment	Continuous assessment by supervisor + Written Report / Thesis + Oral Presentation / Viva
Course Description	The capstone experience of the B.Sc. CS programme. Students may choose one of three pathways: (1) a supervised research or development Project culminating in a written report and oral presentation; (2) an industry Internship of minimum duration as specified by the department, with a reflective report; or (3) a research-based Thesis demonstrating original investigation and academic writing. All pathways require a faculty supervisor and formal assessment. This course maps to all twelve Programme Outcomes.
Suggested Reading	<i>Not applicable — students consult literature relevant to their specific project or thesis topic.</i>

7.2 Selected Programme Elective Courses

CSE 310: Object Oriented Programming (3 Credits)	
Course Code	CSE 310
Credits	3 Credit Hours
Prerequisite(s)	CSE 111 (Hard)
Assessment	Quiz + Assignments + Mid Term Exam + Lab + Final Exam
Course Description	Introduction to object-oriented programming (typically using Java). Topics: object instances; classes; inheritance; polymorphism; abstraction mechanisms; problem domain analysis; system design using an OO approach; OO databases and object persistency; software principles for OO development; advantages and limitations of OO approaches. Includes compulsory 3-hour laboratory each week.
Suggested Reading	<i>Grady Booch, Object-Oriented Analysis and Design (2nd ed., Addison-Wesley, 1993). H. Schildt, Java 2: The Complete Reference (5th ed., McGraw-Hill, 2002).</i>

CSE 419: Programming Languages and Competitive Programming (3 Credits)	
Course Code	CSE 419
Credits	3 Credit Hours
Prerequisite(s)	CSE 111 (Hard), CSE 331 (Hard) or permission of instructor
Assessment	Quiz + Assignments + Mid Term Exam + Final Exam
Course Description	Introduction to the principles of functional, imperative, and logic programming languages. Topics: meta-circular interpreters; semantics (operational and denotational); type systems (polymorphism, inference, and abstract types); object-oriented programming; modules; multiprocessing; and case studies of contemporary programming languages. Programming experience and background in language implementation are expected.
Suggested Reading	<i>Selected readings as designated by the instructor.</i>

CSE 425: Neural Networks (3 Credits)	
Course Code	CSE 425
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz + Assignments + Mid Term Exam + Final Exam
Course Description	An extensive course on neural network architectures and learning algorithms with theory and applications. Topics: temporal and optimal linear associative memories; fuzzy control; Cohen-Grossberg theorem; unsupervised learning; higher-order competitive and differential Hebbian learning; supervised learning; adaptive estimation; stochastic approximation; adaptive vector quantisation; Kohonen self-organising maps; Grossberg theory; simulated annealing; Boltzmann and Cauchy learning; adaptive resonance; Gabor functions and networks.
Suggested Reading	<i>James A. Anderson, An Introduction to Neural Networks (MIT Press, 1995). Mohamad H. Hassoun, Fundamentals of Artificial Neural Networks (MIT Press, 1995).</i>

CSE 427: Machine Learning (3 Credits)

Course Code	CSE 427
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz + Assignments + Mid Term Exam + Final Exam
Course Description	Introduction to machine learning; supervised and reinforcement learning; unsupervised learning algorithms; attribute-based and relational supervised learning algorithms; neural network-based learning algorithms; genetic algorithms and genetic programming; reinforcement learning algorithms; computational learning theory.
Suggested Reading	<i>Tom M. Mitchell, Machine Learning (McGraw-Hill, 1997). Pierre Baldi & Sren Brunak, Bioinformatics: The Machine Learning Approach (2nd ed., MIT Press, 2001).</i>

CSE 428: Image Processing (3 Credits)

Course Code	CSE 428
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz + Assignments + Mid Term Exam + Final Exam
Course Description	Digital image fundamentals; perception and representation; image transforms including FFT, DCT, KLT and Wavelet transform with sub-band decomposition; image enhancement and restoration techniques; image compression techniques; and compression standards (JPEG, MPEG, H.261, H.263).
Suggested Reading	<i>John C. Russ, The Image Processing Handbook (4th ed., CRC Press, 2002). Seul, O'Gorman & Sammon, Practical Algorithms for Image Analysis (Cambridge University Press, 2000).</i>

CSE 431: Natural Language Processing (3 Credits)

Course Code	CSE 431
Credits	3 Credit Hours
Prerequisite(s)	CSE 111 (Hard), CSE 422 (Hard) or consent of instructor
Assessment	Quiz + Assignments + Mid Term Exam + Final Exam
Course Description	Introduction to natural language processing (NLP). Topics: syntax (structure of an utterance); semantics (truth-functional meaning); pragmatics and discourse (context-dependent meaning). Both knowledge-based and statistical NLP methods are introduced, illustrated with text-based and speech-based applications.
Suggested Reading	<i>Daniel Jurafsky & James H. Martin, Speech and Language Processing (1st ed., Prentice Hall, 2000). Christopher D. Manning & Hinrich Schütze, Foundations of Statistical Natural Language Processing (MIT Press, 1999).</i>

CSE 461: Introduction to Robotics (3 Credits)

Course Code	CSE 461
Credits	3 Credit Hours
Prerequisite(s)	CSE 260 (Hard)
Assessment	Quiz + Assignments + Mid Term Exam + Lab + Final Exam

Course Description	Design using MSI and LSI components; memory subsystem design using SRAM and DRAM; design of ALU, memory, and control units (hardwired and microprogrammed); microprocessor-based designs; computer bus standards; design using special-purpose controllers; digital control systems; computers in telecommunication and control. Includes a compulsory 3-hour laboratory each week.
Suggested Reading	<i>R.J. Tocci, N.S. Widmer & G.L. Moss, Digital Systems: Principles and Applications (9th ed., Prentice Hall, 2003).</i>

CSE 471: System Analysis and Design (3 Credits)

Course Code	CSE 471
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz + Assignments + Mid Term Exam + Lab + Project + Final Exam
Course Description	Tools and techniques in systems analysis and design, including data flow diagrams and entity-relationship (E-R) diagrams. Students undertake projects in which they analyse and design a system using these tools. Includes a compulsory 3-hour laboratory each week.
Suggested Reading	<i>Robert J. Thierauf, Systems Analysis and Design of Real-Time Management Information Systems (Prentice Hall, 1975). V. Rajaraman, Analysis and Design of Information Systems (Prentice-Hall India, 2004).</i>

CSE 496: Ethical Hacking (3 Credits)

Course Code	CSE 496
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz + Modular Assignments + Mini Research Project + Presentation + Final Exam
Course Description	Fundamentals of ethical hacking and cybersecurity for students with no prior experience. Provides a balanced mix of theoretical concepts and hands-on practice, focusing on understanding the threat landscape and techniques attackers use to compromise systems. Topics: vulnerability assessment; penetration testing; security awareness; modular assignments on diverse ethical hacking topics. Students complete a mini research project analysing a security problem, proposing solutions, and presenting findings.
Suggested Reading	<i>Stuttard & Pinto, The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws (2nd ed., Wiley, 2011).</i>

7.3 Supporting Mathematics and Physics Courses

MAT 110: Mathematics I: Differential Calculus and Co-ordinate Geometry (3 Credits)

Course Code	MAT 110
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz + Assignments + Mid Term Exam + Final Exam
Course Description	Part 1 — Differential Calculus: Limits; continuity and differentiability; differentiation; Taylor's, Maclaurin's, and Euler's theorems; indeterminate forms; partial differentiation; tangent and normal; subtangent and subnormal; maximum and minimum; radius of curvature and applications. Part 2 — Co-ordinate Geometry: Transformation of co-ordinates; rotation of axes; pairs of straight lines; general equation of second degree; system of circles; conic sections; tangent and normal; asymptotes and applications.
Suggested Reading	<i>Standard Calculus textbook as designated by the instructor.</i>

MAT 120: Mathematics II: Integral Calculus and Differential Equations (3 Credits)

Course Code	MAT 120
Credits	3 Credit Hours
Prerequisite(s)	MAT 110 (Hard)
Assessment	Quiz + Assignment + Lab + Mid Term Exam + Final Exam
Course Description	Integral Calculus: Definitions of integration; methods of substitution, parts, and successive reduction; definite integrals; Walli's formula; improper integrals; Beta and Gamma functions; areas and volumes in Cartesian and polar co-ordinates; arc lengths; surfaces of revolution. Ordinary Differential Equations: Order and degree; formation; solutions of first-order equations; solutions of general linear equations of second and higher order with constant coefficients; homogeneous linear equations; and higher-order equation techniques. Weekly 3-hour tutorial with worksheets.
Suggested Reading	<i>Standard textbook on integral calculus and differential equations as designated by the instructor.</i>

MAT 215: Mathematics III: Complex Variables and Laplace Transformations (3 Credits)

Course Code	MAT 215
Credits	3 Credit Hours
Prerequisite(s)	MAT 216 (Hard)
Assessment	Quiz + Assignments + Mid Term Exam + Final Exam
Course Description	Complex Variables: Complex number systems; functions of a complex variable; limits and continuity; complex differentiation; Cauchy-Riemann equations; mapping by elementary functions; line integrals; Cauchy's integral theorem and formula; Liouville's theorem; Taylor's and Laurent's theorems; singular points; residues; Cauchy's residue theorem; contour integration; conformal mapping. Laplace Transforms: Definitions; transforms of elementary functions; inverse Laplace transforms; transforms of derivatives; unit step and periodic functions; special theorems; solutions of differential equations; evaluation of improper integrals.

Suggested Reading	<i>Standard textbook on complex analysis and Laplace transforms as designated by the instructor.</i>
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MAT 216: Mathematics IV: Linear Algebra and Fourier Analysis (3 Credits)

Course Code	MAT 216
Credits	3 Credit Hours
Prerequisite(s)	MAT 120 (Hard)
Assessment	Quiz + Assignments + Mid Term Exam + Final Exam
Course Description	Linear Algebra: Linear transformations; matrix theory; systems of linear equations; Gaussian elimination; vector spaces and related topics; vector projection; data science applications grounded in linear algebra. Fourier Analysis: Fourier series; Fourier transforms; applications. The course is designed for independent study in four interconnected parts: linear transformations and matrix theory; vector spaces; data science perspectives; and Fourier analysis.
Suggested Reading	<i>Standard textbook on linear algebra and Fourier analysis as designated by the instructor.</i>

STA 201: Elements of Statistics and Probability (3 Credits)

Course Code	STA 201
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz + Assignments + Mid Term Exam + Final Exam
Course Description	Fundamental concepts and tools of statistics and probability. Students develop understanding of statistical and probabilistic methods and their applications in computer science, engineering, natural sciences, economics, and business. Topics: descriptive statistics; probability theory; random variables; common distributions; sampling; estimation; hypothesis testing; regression and correlation.
Suggested Reading	<i>Standard textbook on statistics and probability as designated by the instructor.</i>

PHY 111: Principles of Physics I (3 Credits)

Course Code	PHY 111
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz + Assignment + Lab Reports + Mid Term Exam + Final Exam
Course Description	Principles of Newtonian mechanics and calorimetry. Topics: vectors and scalars; vector addition and subtraction; unit vectors; scalar and vector products; motion in two and three dimensions; projectile motion; uniform circular motion; forces and motion; kinetic energy and work; potential energy and conservation of energy; centre of mass and linear momentum; rotation; rolling; torque and angular momentum; equilibrium and elasticity; gravitation; fluids; and oscillations. Includes 3-hour weekly laboratory experiments.
Suggested Reading	<i>Halliday, Resnick & Walker, Principles of Physics (10th ed., extended). Concepts of Modern Physics by Arthur Beiser.</i>

PHY 112: Principles of Physics II (3 Credits)

Course Code	PHY 112
Credits	3 Credit Hours
Prerequisite(s)	PHY 111 (Hard)
Assessment	Quiz + Assignment + Lab Reports + Mid Term Exam + Final Exam
Course Description	Basic principles of electricity, magnetism, and introductory modern physics. Topics: Coulomb's law; electric field and potential; Gauss's law; electrostatic energy; electric circuits; magnetic field and forces; Faraday's and Lenz's laws; AC circuits; magnetic materials; Maxwell's equations; Rutherford and Bohr atom models; special theory of relativity (time dilation, length contraction); nuclear decays; types of nuclear radiation; and nuclear reactions.
Suggested Reading	<i>Halliday, Resnick & Walker, Principles of Physics (10th ed., extended).</i> <i>Concepts of Modern Physics by Arthur Beiser.</i> <i>Young & Freedman, University Physics with Modern Physics (recommended).</i>

7.4 General Education Course Descriptions

The following courses form part of the University Core (General Education) requirements. All BRAC University students must complete GenEd courses across five streams — Writing Comprehension, Math and Natural Sciences, Arts and Humanities, Social Sciences, and Community Seeking Transformation — to develop broad intellectual competencies alongside technical expertise.

ENG 101: English Fundamentals (3 Credits)

Course Code	ENG 101
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz + Assignments + Participation + Mid Term Exam + Final Exam
Course Description	Designed for students at an upper-intermediate language proficiency level. The course develops all four language skills through two paired strands: speaking and listening, and reading and writing. Students are exposed to academic and professional use of English. Emphasis is placed on expanding vocabulary range, grammatical practice, organisational capacity, and analytical understanding. The course focuses on writing academic papers, analysing reading materials, and developing listening and speaking strategies. Learner-centred and task-based methodologies encourage active participation.
Suggested Reading	<i>Course materials provided by the BIL department.</i>

ENG 102: English Composition I (3 Credits)

Course Code	ENG 102
Credits	3 Credit Hours
Prerequisite(s)	ENG 101 (Hard)
Assessment	Quiz + Assignments + Participation + Mid Term Exam + Final Exam
Course Description	Designed for advanced-level students, this course develops critical reading and academic writing skills. Structured around speaking and listening, and reading and writing strands, it is professional and academic in focus. The course concentrates on academic writing, critical and analytical reading, and formal presentation. Emphasis is placed on developing critical reading and critical thinking abilities through learner-centred and task-based methodologies. Students engage with academic and professional readings and participate actively in classroom activities.
Suggested Reading	<i>Course materials provided by the BIL department.</i>

ENG 103: Advanced Writing Skills and Presentation (3 Credits)

Course Code	ENG 103
Credits	3 Credit Hours
Prerequisite(s)	ENG 102 (Hard)
Assessment	Assignments + Presentations + Mid Term Exam + Final Exam
Course Description	Advanced-level course required only for students who entered the university through ENG 102. Focuses on higher-order academic writing skills and formal presentation techniques. Students develop proficiency in argumentative and research-based writing, and strengthen oral communication skills through structured presentations.
Suggested Reading	<i>Course materials provided by the BIL department.</i>

BNG 103: Bangla Language and Literature (3 Credits)

Course Code	BNG 103
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz + Assignments + Mid Term Exam + Final Exam
Course Description	Develops student proficiency in the Bangla language and builds appreciation for the richness of Bangla literature. Students receive an integrated exposure to Bangla literature through carefully selected essays, short stories, and poetry. The course also covers standard Bangla spelling, joint letters, and correct pronunciation. Students improve their writing and speaking skills in Bangla.
Suggested Reading	<i>Selected literary texts and course materials provided by the department.</i>

HUM 103: Ethics and Culture (3 Credits)

Course Code	HUM 103
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz+ Class Participation+ Presentation + Assignments + Mid Term Exam + Final Exam
Course Description	Introduces students to major theoretical debates and practical applications in ethics. Drawing on classical and modern philosophical perspectives from different historical and geographical settings, the course engages students to identify challenging moral dilemmas, reflect on major ethical traditions and theories, and apply concepts to enduring and real-life ethical problems. The philosophy of ethics examines how we ought to live and act.
Suggested Reading	<i>Selected readings on ethics and moral philosophy as designated by the instructor.</i>

HUM 101: Introduction to World Civilization and Culture (3 Credits)

Course Code	HUM 101
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz+ Class Participation+ Presentation + Assignments + Mid Term Exam + Final Exam

Course Description	Provides an overview of human histories and geographies from the earliest traces of human settlements to approximately 1500 CE. Examines forms of human societies and cultures — cities, states, empires, nomads, tribal confederacies, and land- and sea-based trading network societies. Organised around three themes: earliest agrarian settlements and their nomadic contexts; the role of warfare, trade, religion, migration, and technology in the Afro-Asian and Mediterranean worlds; and the contours of the connected world from the eleventh century onward, focusing on the Indian Ocean and Islamic worlds.
Suggested Reading	<i>Selected historical and cultural readings as designated by the instructor.</i>

EMB 101: Emergence of Bangladesh (3 Credits)

Course Code	EMB 101
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz+ Class Participation+ Presentation + Assignments + Mid Term Exam + Final Exam
Course Description	Designed for students to understand their historic and cultural roots as citizens of Bangladesh. Documents colonial oppression by Britain and (West) Pakistan; analyses political and ethnic subordination, cultural domination, and economic exploitation over the last two centuries. The course traces popular resistance against feudalism and colonialism, the quest for social and economic justice, the development of national consciousness, and the path to independence.
Suggested Reading	<i>Selected historical texts and course materials as designated by the instructor.</i>

POL 101: Introduction to Political Science (3 Credits)

Course Code	POL 101
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz+ Class Participation+ Presentation + Assignments + Mid Term Exam + Final Exam
Course Description	Introduces students to key concepts, approaches, and debates in political science through foundational works in three subfields: political theory, comparative politics, and international relations. Connects important works in the discipline with salient current events, enabling students to engage in critical evaluation of political outcomes and explanatory narratives about politics.
Suggested Reading	<i>Selected readings in political science as designated by the instructor.</i>

CST 305: Borders and Belonging: Past and Future (3 Credits)

Course Code	CST 305
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz+ Class Participation+ Presentation + Assignments + Mid Term Exam + Final Exam
Course Description	Explores how nation-state borders profoundly affect human migration and mobility. Critically analyses the evolving role of territorial identity in shaping citizenship and belonging in an increasingly globalised world. Examines how border-related regulations and political practices have a disproportionate impact on minorities and women. Investigates the consequences of the 1947 Partition, marking the genesis of nation-states in South Asia, and its lasting influence on

	migration patterns.
Suggested Reading	<i>Selected interdisciplinary readings as designated by the instructor.</i>

CST 309: Global Citizenship (3 Credits)

Course Code	CST 309
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Class Participation + Quiz + Assignments + Mid Term Exam + Final Exam
Course Description	Introduces citizenship from historical and conceptual perspectives, exploring its uses and limitations in the global present. The course is divided into two parts. Part One traces the meaning of citizenship in democratic thought; examines bounded concepts of citizenship in modern empires and nation-states; and reviews 'universal' and 'racial' conceptions of membership in the British Empire, before analysing 'nation-state-citizenship' in the twentieth century. Part Two turns to global challenges of statelessness and climate change, focusing on the Rohingya refugee experience and environmental issues in Bangladesh to explore the usefulness and challenges of global citizenship as a concept and practice.
Suggested Reading	<i>Lewis Carroll, Alice in Wonderland (as a framing text); selected academic readings as designated by the instructor.</i>

ECO 101: Introduction to Microeconomics (3 Credits)

Course Code	ECO 101
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz+ Assignments + Mid Term Exam + Final Exam
Course Description	A non-technical introduction to basic microeconomic theory for students from any discipline. Topics: basic concepts in economics; absolute and comparative advantage; gains from specialisation and trade; supply and demand analysis; consumer choice; production and cost analysis; theories of the firm (perfect competition, monopoly, monopolistic competition, oligopoly); theory of distribution and welfare economics; government intervention and policies.
Suggested Reading	<i>Michael Parkin, Melaine Powell & Kent Mathews, Economics (10th ed.). N. Gregory Mankiw, Principles of Economics (10th ed., supplementary).</i>

SOC 101: Introduction to Sociology (3 Credits)

Course Code	SOC 101
Credits	3 Credit Hours
Prerequisite(s)	None
Assessment	Quiz + Assignments + Mid Term Exam + Final Exam
Course Description	An introductory course in sociology for students from any discipline. Familiarises students with key sociological concepts, theories, and theoretical perspectives. Sociology examines social relations, networks, families, groups, cultures, and the rules and procedures that govern modern societies. No prerequisite is required. This course is compulsory for students wishing to declare a minor in sociology.
Suggested Reading	<i>Selected introductory sociology texts as designated by the instructor.</i>

8. Career Opportunities

The B.Sc. CS programme prepares graduates for a wide range of careers in the computing industry, government, academia, and research. Graduates are qualified to implement cutting-edge computer science principles in forward-thinking international organisations. The following career pathways are regularly pursued by BRAC University CSE graduates:

- Software Developer / Software Engineer
- Data Scientist / Data Analyst
- Machine Learning / AI Engineer
- Network Engineer / Systems Administrator
- Database Administrator
- Quality Assurance (QA) Engineer
- IT Consultant / Information Systems Manager
- Multimedia Programmer / Games Developer
- Systems Analyst
- Technical Writer
- Lecturer / Professor (with postgraduate qualification)
- Researcher (Academic or Corporate)

8.1 Graduate Destinations

CSE graduates from BRAC University have been placed at some of the most prestigious technology organisations in Bangladesh and internationally, including:

- Amazon
- Apple
- Google
- Microsoft
- HP
- Samsung R&D
- Grameenphone
- Robi
- Banglalink
- bKash
- BRAC
- Brain Station 23
- BJIT
- Cefalo
- Datasoft
- Dynamic Solution Innovators Ltd
- Enosis Solutions
- NewsCred
- TigerIT
- Therap
- WellDev
- NativeWaves GmbH

9. Further Study and Placement Opportunities

BRAC University maintains a strong track record of graduates successfully pursuing graduate and doctoral studies at internationally recognised universities. The following institutions have accepted BRAC University CSE graduates for further study:

- George Mason University (USA)
- Purdue University (USA)
- University of Delaware (USA)
- Virginia Tech (USA)
- Texas A&M University (USA)
- New York University (USA)
- North Carolina State University (USA)
- University of Washington (USA)
- The California Institute of Technology (USA)
- University of Rochester (USA)
- University of Alberta (Canada)
- The University of British Columbia (Canada)
- University of Toronto (Canada)
- University of Calgary (Canada)
- The University of Sydney (Australia)
- University of Adelaide (Australia)
- Monash University (Australia)

10. Facilities and Support

Computer Vision and Intelligent Systems (CVIS) Lab

The CVIS Research Laboratory inspires innovation among students and faculty. It is dedicated to research in computer vision, data science, artificial intelligence, machine learning, machine vision, medical image processing and analysis, 3D vision, and virtual, augmented and mixed reality.

Thesis Laboratory

Students have access to a dedicated, highly equipped computer laboratory established solely to support thesis and final-year project work, providing the computational resources needed for advanced research.

General Computer Laboratories

Advanced computer laboratories are available across the campus for students to complete coursework, undertake laboratory assignments, and further explore computing concepts beyond the classroom.

Student Tutors / Teaching Assistants

Dedicated tutoring sessions are available 24/7. Student tutors are available on campus from 8:00 am to 5:00 pm. Online support is provided via Discord and Google Meet until 1:00 am.

Faculty Consultation Hours

In addition to weekly lecture and laboratory sessions, all faculty members hold dedicated consultation hours. Students are encouraged to use these sessions for academic guidance, research discussions, and course support.

11. Contact Information

Institution	BRAC University
Department	Computer Science and Engineering (CSE)
Address	Kha 224 Pragati Sarani, Merul Badda, Dhaka 1212, Bangladesh
Telephone	+88 09638464646 Ext. 1001 (Operator)
Email	info@bracu.ac.bd
Website	www.bracu.ac.bd

Certification and Authorisation

This Module Handbook is an official document of the Department of Computer Science and Engineering, BRAC University. It accurately describes the programme structure, course content, and academic regulations applicable to students enrolled in the B.Sc. CS programme.



Signature / Date 06/05/26

Chairperson, Department of CSE

Dr. Sadia Hamid Kazi
Associate Professor and Chairperson
Department of Computer Science and Engineering
BRAC University
Kha 224, Merul Badda, Dhaka- 1212.

